

Quantum Computing — Exam Review

Introduction to Quantum Computing (RWTH Aachen) · exam 20 Aug 2026 · every mistake made across the drill sessions of 18–19 Aug, with the rule that fixes it.

This PDF is for revision only — it cannot go into the exam. The permitted aid is one side of A4, handwritten by you. Printed notes are forbidden and a printout is a disqualification risk. Copy what you want by hand — that copying pass is itself a last retrieval rep. Also forbidden: calculators, phones, any electronic device.

0 · The one that bit twice

×2 · Bernstein-Vazirani: a variable that does not appear in f still occupies a position in s — that bit is 0, and it must be written. $f(x) = x_1 \oplus \neg x_3$ over 3 variables $\rightarrow s = 101$, never “11”. Count the variables in the problem, not the terms in the formula.

1 · Answer-completeness — right maths, lost marks

Historically the most expensive category on this paper. Every one of these was a correct calculation that stopped early or answered the wrong sub-part.

Norm — stopped at the sum of squares

× Reported 6 as the norm of $2|0\rangle + (-1+i)|1\rangle$.

✓ 6 is the squared norm. The norm is $\sqrt{6}$. Sum the $|\cdot|^2$, then take the square root — the last step is a step.

Amplitude vs probability

× For $\psi = [3/5, 4i/5]$ gave 9/25 and 16/25 for all four sub-parts.

✓ Amplitude = the entry verbatim, complex and all (3/5, 4i/5). Probability = $|\text{entry}|^2$ (9/25, 16/25). Amplitude is never squared. These come in 4s — answer all four.

Post-measurement probability

× Gave $P = 1$ for an outcome, reasoning “we already know it happened”.

✓ The probability asked for is the a priori one: the squared norm of the non-normalized post-measurement state, computed from the original amplitudes. Conditioning on the outcome does not make it 1.

Post-measurement state omitted

× Numeric parts answered, the resulting state never written.

✓ Always write the post-measurement state. When both are asked, give non-normalized AND normalized. This single habit is the biggest historical leak.

Multiple x share one f value

× For $f(x) = \lfloor x/2 \rfloor$ with outcome $f=1$, gave only $|10\rangle$.

✓ The collapsed register keeps every x consistent with the measured value — $|10\rangle$ and $|11\rangle$. It is a set, not one term. Check every x , don't stop at the first match.

Stopped past / short of the named point

× Computed through to the end of the circuit instead of the named checkpoint.

✓ The question grades the state at exactly the named point. Apply gates strictly left to right, write the state after each, stop where asked.

Dropped normalization when expanding

× Lost the $1/\sqrt{2}$ factor expanding a tensor product into full ket notation.

✓ $1/\sqrt{2}$ and $2^{-n/2}$ factors survive the whole computation. H on n qubits gives 2^n terms — keep the $2^{-n/2}$.

2 · Complex numbers

Modulus vs modulus squared

× Used $|a+bi|$ where $|a+bi|^2$ was needed — $\sqrt{2}/4 + \sqrt{2}/4$ instead of $1/2 + 1/2$.

✓ $|a+bi|^2 = a^2 + b^2$ (not $(a+bi)^2$, not $\sqrt{(a^2+b^2)}$). $|(1+i)/2|^2 = 2/4 = 1/2$. $|i/\sqrt{2}|^2 = 1/2$, never $-1/2$.

Conjugate vs modulus

× Took the conjugate to mean the modulus $\sqrt{a^2+b^2}$.

✓ Conjugate of $a+bi$ is $a-bi$ — flip the sign of the imaginary part only. Dagger = transpose and conjugate.

Like terms not combined first

× Squared $-|1\rangle$ and $+i|1\rangle$ separately instead of combining to $(-1+i)|1\rangle$.

✓ Collect all coefficients on the same basis ket before squaring anything. $-1+i$ is one amplitude with $|\cdot|^2 = 2$, not two amplitudes.

3 · Indices, ordering and positions

Ket → column vector

✗ Spread one ket's amplitude across several indices, e.g. $|01\rangle \rightarrow [1,0,0,1]$.

✓ Each basis ket maps to exactly one slot. Read the bits as a binary integer = the index. $|01\rangle \rightarrow$ index 1 $\rightarrow [0,1,0,0]$. $|101\rangle$ in 3 qubits is index 5 of 8. Write all 2^n entries, zeros included.

C_{ab} subscript order

✗ Read C_{10} as "both M_1 and M_2 read 1".

✓ Ordered: first digit = M_1 's required outcome, second = M_2 's. C_{10} means $M_1=1$ and $M_2=0$. The I factor is not measured — skip its position entirely when reading the string off.

Tensor product order

✗ Reversed a 3-factor product — placed the last ket first.

✓ $A \otimes B \neq B \otimes A$. Every entry of the first factor scales the whole second factor. Dimensions multiply ($3 \otimes 2 \otimes 2 = 12$), never add.

Separable vs entangled — sign missed

✗ Called $(|00\rangle + |01\rangle + |10\rangle - |11\rangle)/2$ a product state.

✓ Write $[a,b,c,d]$; product iff $ad = bc$. Here $ad = -1/4$, $bc = +1/4 \rightarrow$ entangled. Carry the minus sign into the test — one sign flips the answer.

4 · QFT phase decomposition — three separate slips

$\text{DFT}_{2^n}|x\rangle$ written as a product of $(|0\rangle + e^{i\theta}|1\rangle)/\sqrt{2}$ factors. Two rules, kept apart:

1 · Which bits does a factor use? Leftmost output factor uses the last input bit x_n alone. Each factor rightward adds one more trailing bit. Rightmost uses all bits.

2 · Every factor has its OWN fraction, and every fraction restarts its place values at $1/2$. This is the part that trips: the same bit carries a different weight depending on which factor's fraction it sits in. In the leftmost factor x_n is the first digit, so its weight is $1/2$ — not $1/8$. Only in the rightmost fraction $0.x_1x_2x_3$ does x_3 carry $1/8$.

3 · Then: phase = $2\pi \times$ fraction $\rightarrow$ as a multiple of π , double the fraction.

The whole ladder for $x = 101$ ($x_1=1$, $x_2=0$, $x_3=1$):

Factor	Its own fraction	Value	Phase
leftmost	$0.x_3 = 0.1$	$1/2$	$2\pi \cdot (1/2) = \pi$
middle	$0.x_2x_3 = 0.01$	$0 + 1/4 = 1/4$	$2\pi \cdot (1/4) = \pi/2$
rightmost	$0.x_1x_2x_3 = 0.101$	$1/2 + 0 + 1/8 = 5/8$	$2\pi \cdot (5/8) = 5\pi/4$

Note x_3 is worth $1/2$ in the leftmost factor and $1/8$ in the rightmost — same bit, different fraction, different weight. Reading it as $x_3(1/8)$ everywhere gives $\pi/4$ for the leftmost instead of π .

Used every bit for the leftmost factor

✗ Applied the full formula to the leftmost factor.

✓ Leftmost uses x_n only, and there it sits in the $1/2$ slot: $x=101 \rightarrow 0.x_3 = 0.1 = 1/2 \rightarrow$ phase = π .

Nonzero phase when the bit was 0

✗ Gave a phase for the leftmost factor when $x_n = 0$.

✓ $x_n=0 \rightarrow$ fraction 0 $\rightarrow$ phase 0; that factor is plain $|+\rangle$.

Fraction / 2π arithmetic

✗ Got $7/4$, and separately reported the bare fraction $3/4$ as the answer.

✓ $x=110 \rightarrow 0.110 = 1/2 + 1/4 = 3/4 \rightarrow 2\pi \cdot 3/4 = 3\pi/2$. $x=101 \rightarrow 0.101 = 5/8 \rightarrow 5\pi/4$. Never forget the $\times 2\pi$.

5 · Gate universality — five misses, one checklist

Run both tests, independently. Neither rescues the other.

1 · Entangling gate present? (CNOT, CZ, Toffoli, iSWAP — not SWAP, not any single-qubit gate.) No $\rightarrow$ at best universal for 1-qubit gates.

2 · Do the 1-qubit gates generate a dense subgroup? Fixed gates: need a non-Clifford (T) together with a second axis (H). $\{X,Y,Z\}$, $\{H,S\}$, $\{T\}$ alone, $\{S\}$ alone $\rightarrow$ finite $\rightarrow$ no. Free-parameter rotations $R_x(\theta)$, $R_y(\theta)$ on two axes $\rightarrow$ yes.

Both yes $\rightarrow$ universal. Answer one category, the strongest that applies — never two.

Gate set	Entangling?	Dense 1-qubit gates?	Verdict
{H, S, CNOT}	yes	no — Clifford, finite (24)	not universal
{H, T, CNOT}	yes	yes	universal
{H, S, T, CNOT}	yes	yes	universal
{H, S, T, CNOT, CZ}	yes	yes	universal
{T, CNOT}	yes	no — one axis, cyclic (8)	not universal
{S, CNOT}	yes	no — one axis, cyclic (4)	not universal
{H, T, S} (no CNOT)	no	yes	1-qubit only
{ $R_x(\theta), R_y(\theta)$ }, θ free	no	yes	1-qubit only

Also caught: over-correcting into “small gate sets are never universal” — {H,T,CNOT} is the textbook universal set. Run the two tests; don’t pattern-match on size.

6 · Hamiltonian dynamics — six slips, all mechanical

λ where $e^{-i\lambda t}$ was needed

✗ Wrote the raw eigenvalue into the diagonal of U.

✓ $U = P \cdot e^{-iD_t} \cdot P^\dagger$: the diagonal entries are $e^{-i\lambda t}$, not λ . Exponentiate the diagonal, never entrywise. The minus in e^{-iHt} is load-bearing.

Global phase not factored

✗ Couldn’t pull a scalar out of a diagonal matrix.

✓ $c \cdot \text{diag}(x, y) = \text{diag}(cx, cy)$. A global phase never changes the gate — factor it out before comparing to X, Z, H, S, T.

Solving $2t = \theta$

✗ Skipped or misapplied the divide-by-2.

✓ $t = \theta/2$. Write the equation down, divide, don’t do it in your head.

Euler re-expression treated as a derivation

✗ Thought writing an entry as $e^{i\theta}$ changed its value.

✓ Euler re-expresses the same number in a different form. It transforms nothing.

Gate’s unitary vs action on a state

✗ Brought in an input vector for a “which gate?” question.

✓ “Which unitary does H for time t implement?” needs no input state. Build U, compare to the named gates.

Reused a previous answer

✗ Carried the number 3 from the worked example into a new H and ψ .

✓ Recompute from scratch every time. Energy = $\langle \psi | H \psi \rangle$: compute $H\psi$, then the inner product (conjugate the bra side). The result is a real scalar, not a vector, and not an eigenvalue unless ψ is an eigenvector.

7 · Algorithm-specific

Oracle circuit → f : XOR, not AND

✗ Read accumulated aux terms as ANDed together — gave $x_1 \wedge x_2 \wedge x_3$.

✓ Every gate XORs its contribution onto the target. AND happens only inside one Toffoli’s two controls. CNOT(a→t) adds a; Toffoli(a,b→t) adds $a \wedge b$. Toffoli($x_1, x_2 \rightarrow \text{aux}$) then CNOT($x_3 \rightarrow \text{aux}$) gives $f = (x_1 \wedge x_2) \oplus x_3$. Gates after CNOT(aux→y) uncompute — ignore them.

Shor code — one error counted as two

✗ Block $(|010\rangle + |101\rangle)/\sqrt{2}$ read as two separate bit flips.

✓ A single X on one physical qubit flips that qubit in every term at once. One operator, one location. Sign flip inside a block = phase (Z) error; bit flip inside a triple = X error. Check all three blocks.

Grover — dropped the query factor

✗ Wrote $2^6 \cdot n^3 = n^3$ when combining with oracle cost.

✓ $\sqrt{N}$ queries multiplied by the oracle cost. $N = 2^n \rightarrow 2^{n/2}$; k-bit key $\rightarrow 2^{k/2}$. An $O(n^2)$ oracle scales the total, it does not change the count.

Bomb tester — jumped to 50/50

✗ Gave $P(D_0)=P(D_1)=1/2$ and $P(\text{explode})=0$ with the bomb present.

✓ The bomb destroys interference. Branch first: $P(\text{explode}) = P(\text{path A})$. What survives then splits at the second beam splitter. 50/50 first splitter $\rightarrow 1/2, 1/4, 1/4$. The three must sum to 1.

Sub-state vs probability

✗ Isolated the relevant terms and called that the probability.

✓ A sub-state is a vector; a probability is a scalar. Sum the squared magnitudes. “Without knowing the first result” = marginalize; “knowing it” = collapse and renormalize first, then square.

BV circuit modifications — the tempting wrong call

✗ Treating “the outcome is ignored” as “the measurement is harmless”.

✓ Measuring the x-register before U_f destroys the superposition → breaks it. Measuring and discarding the ancilla after U_f is harmless — it stays in $|-\rangle$, unentangled from x. An ignored outcome still collapses; unentangled is what makes it safe.

8 • Fast checks worth having cold

Valid quantum state?	$\sum a_i ^2 = 1$ exactly. $ a+bi ^2 = a^2+b^2$.
Valid probability vector?	all entries ≥ 0 , entries sum to 1 (not squares).
Valid stochastic matrix?	no negative entry (check first), then every column sums to 1 — columns, not rows.
Valid unitary?	$U^\dagger U = I$; fast route — columns orthonormal. Dagger = transpose and conjugate.
Entangled (2 qubits)?	$[a,b,c,d]$: product iff $ad = bc$.
Partial-measurement M ?	diagonal 0/1 projector: 1 on consistent basis states. $M^2 = M$, $M^\dagger = M$, not unitary. $P = \ M\psi\ ^2$.
$ e^{i\theta} = ?$	1, always, for real θ . The binary fraction is a distractor.
DFT period?	input comb spacing r on N points → output period N/r . Inverse, not equal.
Uniform superposition?	every basis state, equal amplitude and equal sign. $H^2 = I$ (H twice = back to $ 0\rangle$). X never creates superposition. H then CNOT = Bell, not uniform.
QEC below threshold?	multiply p by the physical gate count first, then compare to p_{th} .
Shor factor extraction	r odd or $a^{r/2} \equiv -1 \pmod N \rightarrow$ attempt fails, say so. Else $\gcd(a^{r/2} \mp 1, N)$, report the non-trivial one.
Photon energy	the difference between levels, never the absolute value.
Stabilizers under U	conjugate: $S \rightarrow U S U^\dagger$ (not $U S$). H : $X \leftrightarrow Z$, $Y \rightarrow -Y$. S : $X \rightarrow Y$. Carry every sign.

9 • Sixty seconds before you start writing

- Read what the question asks for — amplitude or probability, normalized or not, state or number, which specific checkpoint.
- Answer every sub-part. Incomplete answers, not wrong maths, cost the most marks in every session.
- Squaring: probability yes, amplitude no. Norm: square root at the end, yes.
- Signs and $1/\sqrt{2}$ factors survive to the last line.
- Order matters — tensor factors, C_{ab} subscripts, basis indices, binary fractions.
- Answers are typed into Dynexite in fixed syntax: $\text{ket}(01)$, $\text{sqrt}(2)$, $1/\text{sqrt}(2)$, matrices as stacked boxes. Partial credit is stated per sub-part — a non-normalized answer still scores, so write something.

32 logged mistakes from 79 drills across 18–19 Aug 2026. Every archetype that appeared on the real 2024 paper is drilled to exam-ready. Revision aid only — the exam sheet must be handwritten.